



What goes along with a student's final grade?

Findings from real school data, for a school leader deciding where to focus student support

[Team name] · [Presenter names]

CONTEXT

The problem we are solving

A school has limited time, budget and staff for student support, so it has to choose who to help first.



Who decides

A principal or student-support lead



The decision

Where to direct tutoring and support first



What we offer

Evidence on which factors go along with final grades

CONTEXT

Our question

AIM

Find which student, family and lifestyle factors are most strongly associated with final grade (G3).

G3 is the final grade, scored out of 20.

1

Which factors (study time, past failures, absences, family) go with higher or lower grades?

2

Do alcohol use and going out go with lower grades, after allowing for study time?

3

Does school and family support go with better grades, or flag students already struggling?

DATA

The data

Group	Examples of what is recorded
Student	age, sex, absences, past class failures
Study and school	study time, school support, wants higher education
Family	parents' education, family support, family relationships
Lifestyle	free time, going out, alcohol use, health
Outcome	G3, the final grade (0 to 20)

33

attributes recorded per student

0 to 20

scale of the final grade, G3

649

students (Portuguese course file)

DATA

Preparing the data

Checked for gaps and duplicates

Result: 0 duplicate rows and 0 missing values

Left out earlier grades

G1 and G2 are earlier grades that almost give away the final grade (G3), so we excluded them.

Checked zero grades

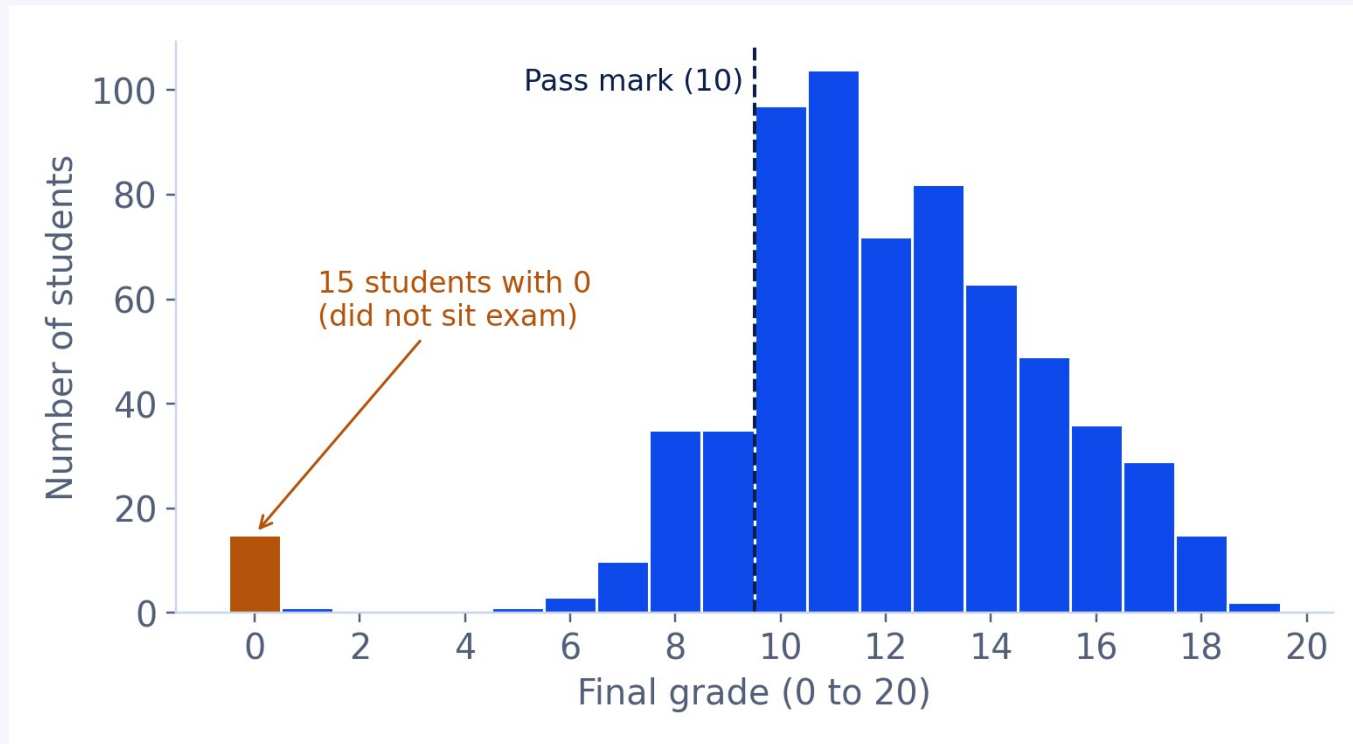
15 students had $G3 = 0$ with no absences (did not sit the exam), so we left them out: 634 remain

Handled skewed answers

Absences are skewed (most 0 to 6, max 32); kept as counts and checked no student drives results

EXPLORATION

How final grades are spread



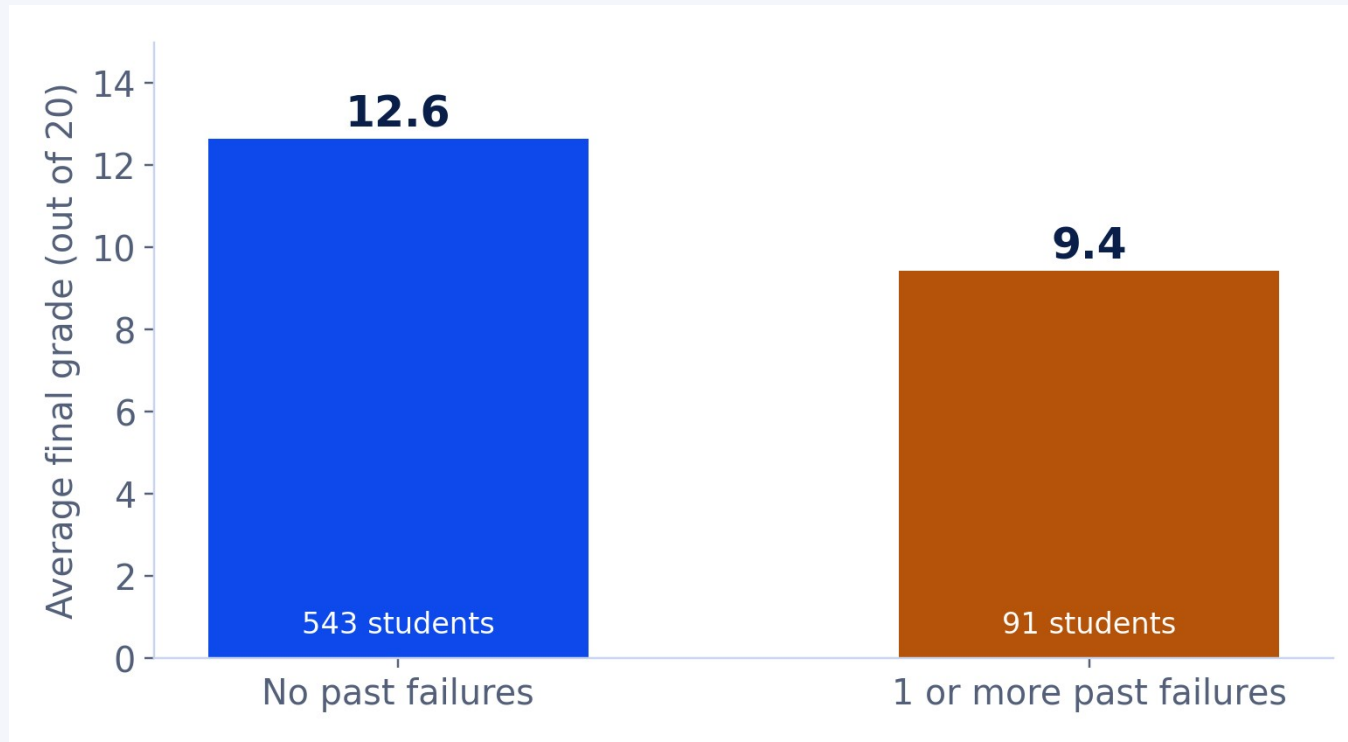
KEY TAKEAWAY

Most students score 10 to 14; a typical grade is 12.

15 zeros = students who did not sit the exam

EXPLORATION

Study habits versus grades



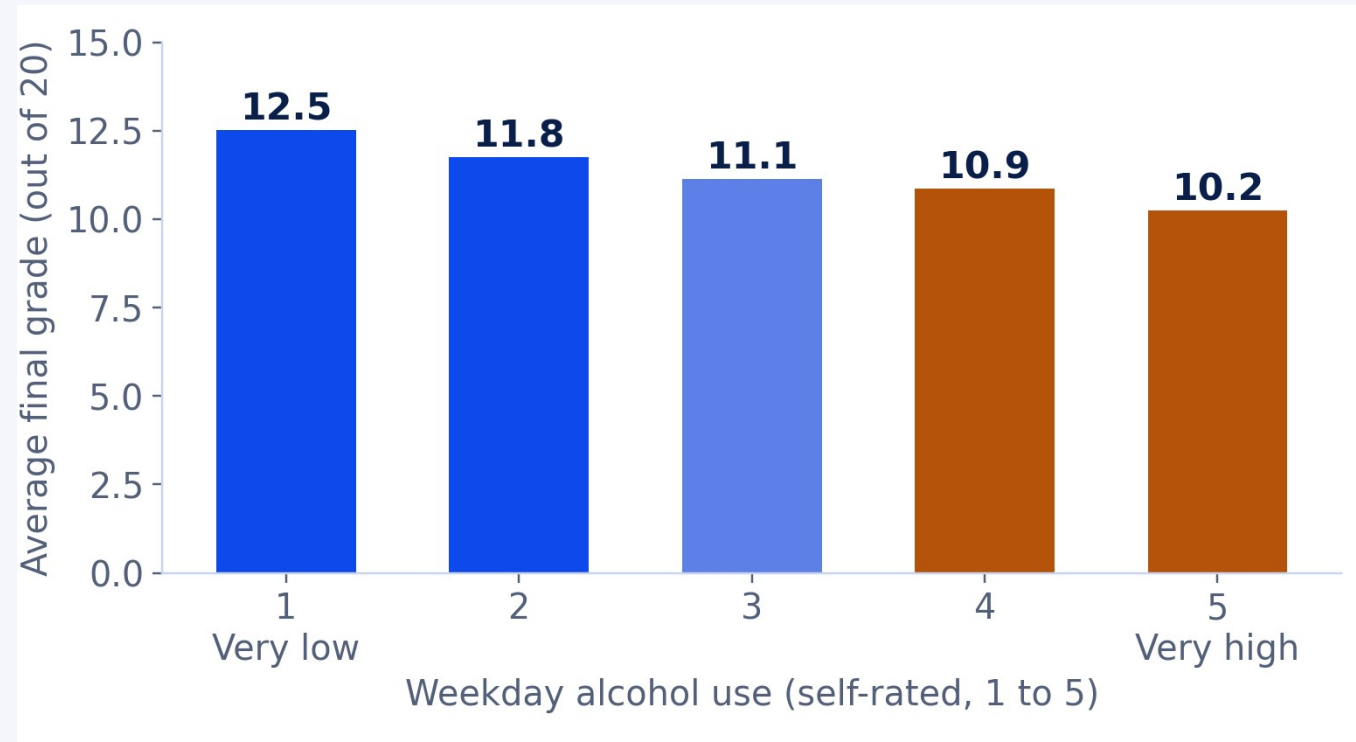
KEY TAKEAWAY

Past failures: about 3 points lower (9.4 vs 12.6).

Links to Question 1: yes, the strongest warning sign.

EXPLORATION

Lifestyle versus grades



KEY TAKEAWAY

More weekday alcohol, lower grades: 12.5 down to 10.2.

Links to Question 2. This shows association only.

MODEL

The model

What goes in

Study time, past failures, absences, family and lifestyle answers



The model

Linear regression compares students who are alike in everything else



What comes out

How many grade points each factor goes with

HOW WELL DOES IT FIT?

$$R^2 = 0.36$$

the share of differences in grades the model explains

ASSUMPTIONS WE CHECKED

Residuals near normal, no outlier drives results, little overlap ($VIF \leq 1.2$); used robust errors

MODEL

What we found

FINDING 1

Past failures

–1.3 points for each past failure
(study time: +0.3 per level)

FINDING 2

Lifestyle

–0.2 per level of alcohol or going
out, after study time

FINDING 3

Support

School support –1.1 (flags
strugglers); family: no link

Association, not cause: these factors go along with grades, but we cannot say they cause them.

MEANING

What this means for a school leader

Each past failure: about 1.3 points lower



Offer early tutoring to students who have failed before

Alcohol and going out: slightly lower grades



Add lifestyle advice to wellbeing programs

Supported students still score 1.1 lower



Re-check the support list after term 1

These are options supported by association, not proof that an action will raise grades.

LIMITS

Limitations

Limitation	What it means for our conclusion
Sample of 649 students (634 in the model)	Estimates are imprecise; small gaps may be noise
Two Portuguese schools, about 2005 to 2006	May not apply to Australian students today
Self-reported 1 to 5 ratings	Answers may be biased or inconsistent
Observational data	We show association, not cause
Earlier grades (G1, G2) left out	We explain grades from background, not past results

The data is about minors, so we use it only to look at group patterns.

TAKE-HOME MESSAGE

Past failures, study habits and lifestyle go along with final grades, so support is best directed early at students who have failed before.

Thank you.
Questions?

[Team name] · [Member names]